

# Seizure outcomes and mesial resection volumes following selective amygdalohippocampectomy and temporal lobectomy

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**Object.** Anterior temporal lobectomy (ATL) and selective amygdalohippocampectomy (SelAH) are the preferred surgical approaches for the treatment of medically refractory epilepsy involving the nondominant and dominant temporal lobes, respectively. Both techniques provide access to mesial structures—with the ATL providing a wider surgical corridor than SelAH. Because the extent of mesial temporal resection potentially impacts seizure outcome, the authors examined mesial resection volumes, seizure outcomes, and neuropsychiatric test scores in patients undergoing either ATL or transcortical SelAH at a single institution.

**Methods.** A retrospective study was conducted in 96 patients with medically refractory mesial temporal lobe epilepsy. Fifty-one patients who had nondominant temporal lobe epilepsy underwent standard ATL, and 45 patients with language-dominant temporal lobe epilepsy underwent transcortical SelAH. Volumetric MRI analysis was used to quantify the mesial resection in both groups. In addition, the authors examined seizure outcomes and the change in neuropsychiatric test scores.

**Results.** Seizure-free outcome in the entire patient cohort was 94% at a mean follow-up of 44 months. There was no significant difference in the seizure outcome between the 2 groups. The extent of resection of the mesial structures following ATL was slightly higher than for SelAH (98% vs 91%,  $p < 0.0001$ ). The change in neuropsychiatric test scores largely reflected the side of surgery, but overall IQ and memory function did not change significantly in either group.

**Conclusions.** Transcortical SelAH provides adequate access to the mesial structures, and allows for a resection that is nearly as extensive as that achieved with standard ATL. Seizure outcomes and neuropsychiatric sequelae are similar in both procedures.

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**KEY WORDS** • epilepsy surgery • temporal lobe • anterior temporal lobectomy • amygdala • hippocampus • entorhinal cortex

THE treatment of medically refractory mesial temporal lobe epilepsy is predominantly surgical, and the results of the ATL have been well described.<sup>5,6,14,16,22</sup> Seizure-free outcomes usually fall in the range of 60%–70%, although higher outcomes have been reported.<sup>12,15,19</sup> The extent of mesial resection necessary for optimal seizure outcome following ATL has been investigated in a number of studies. In 1984, a study published by Spencer et al.<sup>17</sup> suggested that 20% of patients undergoing ATL

have the primary epileptogenic focus within the posterior hippocampus. This has prompted some to suggest that postoperative seizure outcome may be improved with a more complete hippocampal resection. A prospective, blinded study in which 70 patients were randomized into 2 groups differing in the extent of hippocampal resection showed that patients undergoing complete hippocampal resections had better postoperative seizure outcomes than those undergoing conservative resections (69% vs 38% seizure free at 1 year).<sup>24</sup> This study and other retrospective series have suggested that, when it comes to hippocampal resection, more is better.<sup>2,9</sup>

Whereas a radical excision of the hippocampus is not typically problematic in nondominant temporal lobe

*Abbreviations used in this paper:* ATL = anterior temporal lobectomy; BNT = Boston Naming Test; FSIQ = full-scale IQ; MMSE = Mini-Mental State Examination; PIQ = performance IQ; PM = pictorial memory; SelAH = selective amygdalohippocampectomy; VIQ = verbal IQ; VM = verbal memory.

surgery, resections in the language-dominant hemisphere are influenced by the presence of important language-bearing regions within the temporal neocortex. Broad neocortical resection is therefore not usually advocated unless the regions resected are devoid of language function. This approach has led some to advocate resections either guided by intraoperative stimulation mapping or extraoperative mapping with subdural grids. Although either of these approaches allows for broad exposure of the mesial temporal lobe, the former involves a lengthy and somewhat stressful operation, and the latter requires 2 separate operations and a lengthy hospital stay. Maximal neocortical exposure therefore involves added risk, patient inconvenience, and costs.

Other surgical techniques used in dominant temporal lobe resection take a less radical approach. The superior temporal gyrus-sparing temporal lobectomy, for example, is based on the assumption that most temporal lobe language areas reside in the superior temporal gyrus.<sup>18</sup> This approach affords extensive mesial exposure with a lower risk of postoperative language deficit. A more minimally invasive surgical technique is the SelAH. Pioneered by Niemeyer<sup>10</sup> in the 1950s, and Wieser and Yaşargil<sup>23</sup> in the 1970s, SelAH minimizes lateral temporal injury during removal of the mesial structures. Yaşargil's original approach involved a transsylvian exposure, which involves fairly extensive dissection of the middle cerebral artery branches.<sup>26</sup> The risks of vascular injury and retraction injury have been blamed in the reports of stroke and language deficit associated with this technique.<sup>1,28</sup> With the advent of image guidance surgery, a transcortical SelAH involving a corticotomy of the middle temporal gyrus has offered a safer approach to the mesial structures.<sup>11,21</sup> Finally, a subtemporal SelAH has been described, which further reduces the risk of temporal lobe intrusion while minimizing the risk to the middle cerebral artery.<sup>7</sup>

The various merits and disadvantages of these approaches have been argued in the literature, but comparisons have been difficult due to the lack of data from randomized studies. Still, the overarching concern regarding minimally invasive approaches has centered on the reduced access corridor to the mesial temporal lobe, which could theoretically impact seizure outcome.<sup>13,17,24</sup> Several groups have nevertheless examined the seizure outcomes following SelAH, and have reported outcomes on par with standard anteromesial resections reported in the literature.<sup>3,12,19</sup> However, there is as of yet no direct comparison of mesial resection volumes between SelAH and standard ATL. We set out to determine whether there is a difference in the extent of mesial resection between SelAH and standard ATL in a single institution, and whether there are any differences in seizure outcomes or in neuropsychological outcome.

## Methods

### *Patient Population*

We retrospectively identified all patients who underwent temporal lobe resections for mesial temporal epilepsy at the University of Michigan between 2001 and

2007. All patients underwent an extensive presurgical workup that included interictal electroencephalographic recordings, neuropsychological evaluation, and speech and language testing, as well as video electroencephalographic monitoring, imaging studies, and intracarotid methohexital testing (Wada test). During this time period, all nondominant resections consisted of a standard ATL, according to techniques described elsewhere.<sup>17,18</sup> Resections of the mesial temporal structures in the language-bearing hemisphere were all performed using an image-guided, transcortical SelAH, according to established techniques.<sup>11,25,27</sup> All operations were performed by a single surgeon (O.S.). Patients with pathological entities suspected outside the mesial temporal lobe were excluded from the analysis. Outpatient clinic notes from both neurosurgical and neuroepilepsy follow-up visits were reviewed. Demographic data, duration of epilepsy, side of surgery, surgical technique, use of invasive monitoring, and pathological entity were recorded.

We examined the seizure outcomes of these patients according to the Engel Classification (Table 1) at various follow-up intervals.<sup>4</sup> At each point, we considered the occurrence of seizures in the prior 1-year period when determining the seizure outcome class (unless less than 1 year had elapsed since surgery at that point). In addition, the volumes of resected tissue in the mesial temporal lobe were determined in the regions of the amygdala, hippocampus, and entorhinal cortex (see below). Finally, neuropsychological test data (VIQ, PIQ, FSIQ, BNT, VM, PM, and MMSE) in these patients were analyzed preoperatively and again 3 months postsurgery. A minimum of 1 year of follow-up seizure data were available for all patients included in the analysis. Patients with shorter follow-ups were excluded.

### *Volumetric Analysis With MRI*

Volumetric MRIs were acquired using a 1.5-T scanner (GE Healthcare or Phillips Medical Systems). High-resolution spoiled gradient-recalled acquisition sequences were acquired using a head coil at 1.5- to 1.7-mm-thick slices with no interslice gap. Preoperative and postoperative residual volumetric measurements following temporal lobe surgery were obtained in the following manner: a single operator (J.P.T.) performed manual tracings around respective mesial temporal structures as viewed on coronal slices to form 2D areas. The traced areas of these cross-sections were calculated in square millimeters by our MRI software package (Advantage Windows, AW 4.3; GE Healthcare). Multiplication of the sum of cross-sectional areas by the image slice thickness yielded a volumetric measurement. Subtracting the residual postoperative volumetric measurement from that obtained preoperatively yielded the volume resected; dividing this volume by the preoperative volumetric measurement yielded the extent of resection (Fig. 1). To maintain consistency, volumetric measurements of the amygdala were made caudal to the anterior commissure, and hippocampal measurements were made rostral to the posterior aspect of the collicular plate. Randomly selected studies were analyzed by another author (D.G.H.) for quality assurance. The anatomical guidelines used for the identi-

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**TABLE 1: Engel classification scheme**

| Class            | Definition                                                                                                    |
|------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Class I</b>   | <b>seizure free*</b>                                                                                          |
| A                | completely seizure free since surgery                                                                         |
| B                | aura only since surgery                                                                                       |
| C                | some seizures after surgery, but seizure free for at least 2 yrs                                              |
| D                | atypical generalized convulsion w/ antiepileptic drug withdrawal only                                         |
| <b>Class II</b>  | <b>rare seizures ("almost seizure-free")</b>                                                                  |
| A                | initially seizure free, but has rare seizures now                                                             |
| B                | rare seizures since surgery                                                                                   |
| C                | more than rare seizures after surgery, but rare seizures for at least 2 yrs                                   |
| D                | nocturnal seizures only, which cause no disability                                                            |
| <b>Class III</b> | <b>worthwhile improvement</b>                                                                                 |
| A                | worthwhile seizure reduction                                                                                  |
| B                | prolonged seizure-free intervals amounting to greater than half the follow-up period, but not less than 2 yrs |
| <b>Class IV</b>  | <b>no worthwhile improvement</b>                                                                              |
| A                | significant seizure reduction                                                                                 |
| B                | no appreciable change                                                                                         |
| C                | seizures worse                                                                                                |

\* Excludes early postoperative seizures (first few weeks).

fication of the hippocampus, amygdala, and entorhinal cortex have been previously described.<sup>8,20</sup>

### Statistical Analysis

Comparisons of means and calculations of signifi-

cance were performed using ANOVA through the JMP statistical software package (JMP v.8, SAS Institute). Modeling by generalized estimating equations analysis was performed using STATA v.10 (STATA Corp.). Patient data were clustered according to follow-up, and seizure outcome (family variable) was set as binomial. Correlations were performed in an exchangeable fashion.

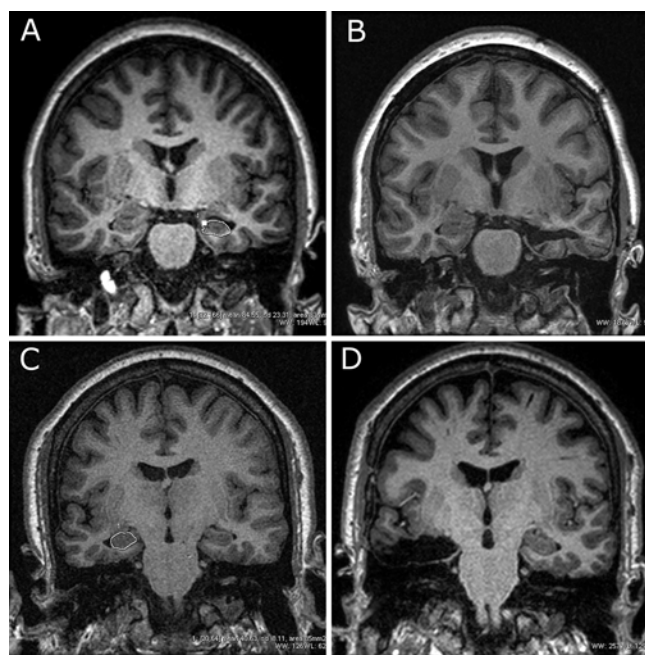
## Results

### Patient Characteristics

Between 2001 and 2007, 106 patients underwent temporal lobe surgery for epilepsy. Ten of these patients had follow-up periods of less than 1 year and were excluded from the analysis. Of the remaining 96 patients, 51 underwent ATL and 45 underwent SelAH procedures following presurgical evaluation. Of the patients undergoing ATL, all operations were performed on the nondominant hemisphere (74.5% right; 25.5% left). All patients undergoing SelAH had operations performed on the (language-bearing) left hemisphere. Craniotomy procedures followed by placement of subdural grids and/or depth electrodes were required for initial localization of seizure foci in 35.6% of patients undergoing SelAH and in 17.6% of patients undergoing ATL. The mean duration of follow-up was 43.2 months (ATL group) and 44.7 months (SelAH group). All patients were followed for at least 1 year. Further details are included in Table 2.

### Volumetric Analysis

Anterior temporal lobectomy allowed for a near-total resection of the amygdala (99.5%), whereas the extent of resection in SelAH was slightly but significantly lower (93.7%,  $p < 0.0001$ ). Similarly, the extent of resection of



**FIG. 1.** Examples of coronal MRIs showing the volumetric calculation method. **A:** Preoperative SelAH. **B:** Postoperative SelAH. **C:** Preoperative ATL. **D:** Postoperative ATL. Examples of area tracings outlining the hippocampus are demonstrated on preoperative images. Volumes of mesial temporal structures were calculated based on sequential area tracings and imaging slice thickness.

**TABLE 2: Characteristics in 96 patients with medically refractory, surgically treated epilepsy\***

| Characteristic                               | SelAH            | ATL              |
|----------------------------------------------|------------------|------------------|
| no. of patients                              | 45               | 51               |
| mean FU in mos (range)                       | 44.7 (12.3–96.1) | 43.2 (13.3–91.5) |
| sex                                          |                  |                  |
| M                                            | 20 (44.4%)       | 26 (51.0%)       |
| F                                            | 25 (55.6%)       | 25 (49.0%)       |
| handedness                                   |                  |                  |
| rt                                           | 36 (80%)         | 43 (84.3%)       |
| lt                                           | 8 (17.8%)        | 6 (11.8%)        |
| ambidextrous                                 | 1 (2.2%)         | 2 (3.9%)         |
| epilepsy                                     |                  |                  |
| mean age at onset of epilepsy in yrs (range) | 17.1 (1–47)      | 15.0 (1–52)      |
| mean epilepsy duration in yrs (range)        | 21.2 (1–42)      | 21.0 (1–57)      |
| w/ secondary generalization                  | 31 (68.9%)       | 35 (68.6%)       |
| w/o secondary generalization                 | 14 (31.1%)       | 16 (31.4%)       |
| pathological entity                          |                  |                  |
| MTS                                          | 37 (82.2%)       | 41 (80.4%)       |
| gliosis                                      | 8 (17.8%)        | 0                |
| dysplasia                                    | 2 (4.4%)         | 8 (15.7%)        |
| cavernous malformation                       | 1 (2.2%)         | 0                |
| hamartoma                                    | 0                | 2 (3.9%)         |
| >1 pathology                                 | 3 (6.7%)         | 4 (7.8%)         |
| op data                                      |                  |                  |
| mean age at op in yrs (range)                | 38.3 (17–57)     | 36.0 (19–59)     |
| side of op                                   |                  |                  |
| rt                                           | 0                | 38 (74.5%)       |
| lt                                           | 45 (100%)        | 13 (25.5%)       |
| w/ grids                                     | 16 (35.6%)       | 9 (17.6%)        |
| w/o grids                                    | 29 (64.4%)       | 42 (82.4%)       |

\* FU = follow-up; MTS = mesial temporal sclerosis.

the hippocampus during ATL was higher than in SelAH (95.8% vs 89.2%,  $p < 0.0001$ ). Finally, resection of the entorhinal cortex was more complete during ATL than for SelAH (100.0% vs 89.8%,  $p < 0.0001$ ). The total mesial resection was therefore more complete in the ATL than in the SelAH group. These differences are illustrated in Fig. 2.

#### Seizure Outcomes

**The ATL Group.** Three months after surgery, 47 patients (92.1%) were seizure free (Class I). At 1 year postoperatively, 8 patients (15.7%) who were initially seizure free developed rare seizures (Class II). One patient improved (from Class III to Class II). At 2 years postoperatively, 3 patients improved to Class I and 2 patients worsened; one of these developed rare seizures (Class II) and the other developed seizures with notable worthwhile improvement (Class III). Seven patients were lost to follow-up. Three years after surgery, 85.2% of patients

who had undergone ATL were seizure free (Class I). Two patients improved (both Class II; to Class I) and 1 patient developed rare seizures (from Class I to Class II). Seven patients were either lost to follow-up or did not have the requisite follow-up periods. These results are shown in Table 3. At last follow-up, 92.2% of all patients who had undergone ATL had Class I outcomes, with a mean follow-up of 43.2 months (range 13.3–91.5 months). Of the 44 patients with follow-up periods of at least 2 years, 38 were seizure free (Class I) at that point; of these 38 patients, and 37 were seizure free at 1 year.

**The SelAH Group.** Three months following SelAH, 42 patients (93.3%) were seizure free (Class I). At 1 year postoperatively, 4 patients who were initially seizure free developed rare seizures (Class II), and 1 patient improved from Class III to Class I. At 2 years postoperatively, 4 patients improved to Class I from Class II. Six patients were either lost to follow-up or did not have adequate periods of follow-up. At 3 years postoperatively, 93.1% of patients who had undergone SelAH were seizure free (Class I). Four patients improved to Class I (all from Class II), and 16 patients were unable to meet this period of follow-up. These results are presented in Table 3. At last follow-up, 95.6% of all patients who had undergone SelAH had a Class I outcome, with a mean follow-up of 44.7 months (range 12.3–96.1 months). Of the 39 patients with follow-up periods of at least 2 years, 35 were seizure free (Class I) at that point; of these 35 patients, 32 were seizure free at 1 year.

#### Factors Predictive of Seizure Outcome

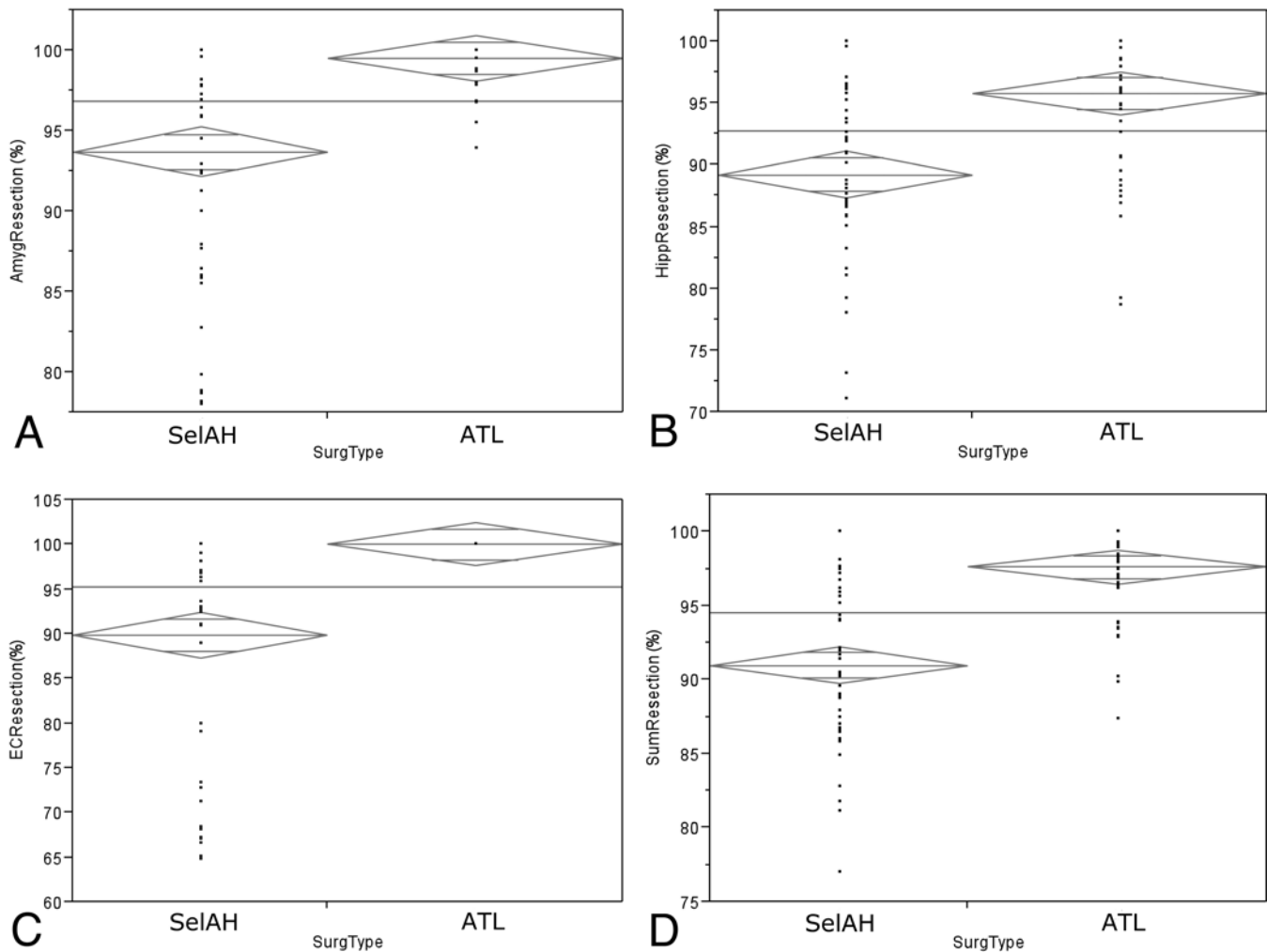
Modeling with generalized estimating equations was used to determine the association of seizure outcome in the Engel class, with several factors as listed in Table 4. Data were clustered according to each patient's progress through his or her own unique period of follow-up, and outcomes were grouped in a binomial fashion—as either Class I (favorable) or non-Class I (unfavorable). Based on

**TABLE 3: Seizure outcomes following operation\***

| Op    | Engel Class | No. (%) at FU |           |           |           |
|-------|-------------|---------------|-----------|-----------|-----------|
|       |             | 3 Mos         | 1 Yr      | 2 Yrs     | 3 Yrs     |
| SelAH |             |               |           |           |           |
|       | I           | 42 (93.3)     | 38 (84.4) | 35 (89.7) | 27 (93.1) |
|       | II          | 2 (4.4)       | 7 (15.6)  | 4 (10.3)  | 2 (6.9)   |
|       | III         | 1 (2.2)       | 0         | 0         | 0         |
|       | IV          | 0             | 0         | 0         | 0         |
| total |             | 45            | 45        | 39        | 29        |
| ATL   |             |               |           |           |           |
|       | I           | 47 (92.1)     | 43 (84.3) | 38 (86.4) | 23 (85.2) |
|       | II          | 3 (5.9)       | 8 (15.7)  | 5 (11.4)  | 3 (11.1)  |
|       | III         | 1 (2.0)       | 0         | 1 (2.2)   | 1 (3.7)   |
|       | IV          | 0             | 0         | 0         | 0         |
| total |             | 51            | 51        | 44        | 27        |

\* Data are given as the number of patients grouped by Engel class at points of follow-up.

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**Fig. 2.** Graphs showing comparisons of percent resection, grouped by operation type. **A:** Amygdala (Amyg). **B:** Hippocampus (Hipp). **C:** Entorhinal cortex (EC). **D:** Sum of mesial temporal structures. The group mean is reflected in the horizontal line across each means diamond. Resection volumes were higher in the ATL group in all mesial structures ( $p < 0.0001$ ). The apex and base of each diamond delimit the 95% confidence interval. The mean of the entire patient population is shown in the horizontal line across the whole graph.

the generated model, the implantation of grids and/or depth electrodes was shown to be weakly associated with unfavorable seizure outcomes ( $p = 0.05$ ). Notably, the type of operation (SelAH or ATL) and the extent of resection of mesial temporal structures did not correlate with a favorable seizure outcome over time. Sex, patient age at operation, handedness, preoperative secondary generalization, and the duration of a patient's epilepsy similarly do not appear to correlate with seizure outcome over time.

### Neuropsychiatric Test Outcomes

**The BNT.** Expressive language function, as measured by the BNT, did not change significantly with either ATL or SelAH. Preoperative and postoperative BNT scores in patients undergoing ATL differed only slightly (preoperative 49.9, postoperative 49.0 [ $p = 0.650$ ]). Patients undergoing SelAH experienced slightly larger deficits in BNT scores; nevertheless, these changes also failed to achieve statistical significance (preoperative 45.0, postoperative

41.5 [ $p = 0.078$ ]). These results are represented graphically in Fig. 3A.

**Intelligence Quotient Testing: VIQ, PIQ, and FSIQ.** The VIQ scores remained stable between the preoperative state and the 3-month postoperative evaluation in the SelAH cohort (preoperative 86.1, postoperative 85.5 [ $p = 0.825$ ]). Patients undergoing ATL procedures had a slight increase in VIQ scores, although this too was not statistically significant (preoperative 89.7, postoperative 91.3 [ $p = 0.675$ ]). Both the ATL (preoperative 91.4, postoperative 97.3 [ $p = 0.158$ ]) and SelAH cohorts (preoperative 91.5, postoperative 95.0 [ $p = 0.313$ ]) demonstrated statistically insignificant increases in PIQ scores. Similar increases were demonstrated with FSIQ scores (ATL: preoperative 90.5, postoperative 94.3 [ $p = 0.309$ ]; SelAH: preoperative 88.3, postoperative 89.4 [ $p = 0.682$ ]). These results are shown in Fig. 3D.

**Memory Testing: VM and PM.** Patients undergoing SelAH experienced a statistically significant decrease in

**TABLE 4: Generalized estimating equations modeling of parameters having a possible influence on favorable seizure outcome**

| Parameter                | Coefficient | SEM   | p Value |
|--------------------------|-------------|-------|---------|
| type of op (ATL)         | 0.088       | 0.071 | 0.22    |
| age at op                | -0.162      | 0.257 | 0.53    |
| male sex                 | 0.051       | 0.053 | 0.33    |
| rt-handedness            | -0.080      | 0.075 | 0.28    |
| secondary generalization | 0.049       | 0.058 | 0.39    |
| age at onset of epilepsy | 0.162       | 0.258 | 0.53    |
| duration of epilepsy     | 0.160       | 0.258 | 0.53    |
| dual pathology           | -0.002      | 0.100 | 0.98    |
| grid implantation        | 0.124       | 0.064 | 0.05*   |
| complication             | 0.028       | 0.075 | 0.71    |
| extent of resection      |             |       |         |
| % sum                    | -0.032      | 0.036 | 0.37    |
| % amygdala               | 0.018       | 0.014 | 0.21    |
| % hippocampus            | 0.012       | 0.019 | 0.52    |
| % entorhinal cortex      | 0.001       | 0.005 | 0.87    |

\* Statistically significant.

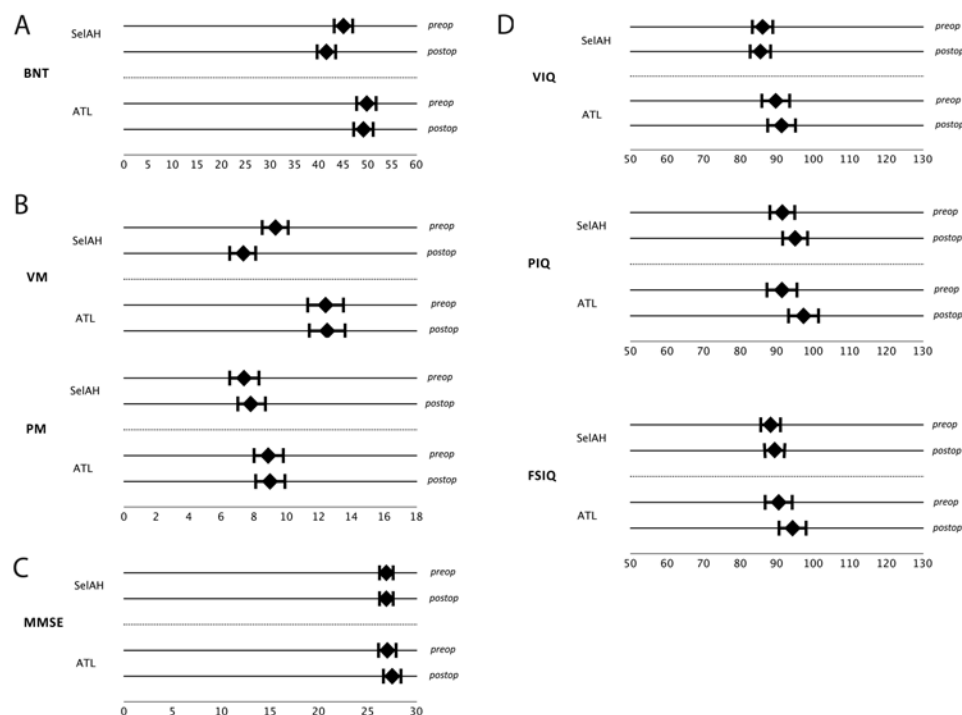
VM scores following surgery (preoperative 9.32, postoperative 7.34 [ $p = 0.013$ ]). The VM scores remained stable in the ATL group (preoperative 12.4, postoperative 12.5 [ $p = 0.974$ ]). Patients undergoing either procedure experienced no significant change in PM score (ATL: preoperative 8.87, postoperative 8.98 [ $p = 0.90$ ]; SelAH: preopera-

tive 7.38, postoperative 7.79 [ $p = 0.658$ ]). These results are illustrated in Fig. 3B.

*The MMSE.* Overall cognitive performance did not change in patients undergoing either ATL or SelAH. Preoperative and postoperative scores on the MMSE demonstrated no real change in either group (ATL: preoperative 27.0, postoperative 27.5 [ $p = 0.618$ ]; SelAH: preoperative 26.9, postoperative 26.9 [ $p = 0.933$ ]). This finding is shown graphically in Fig. 3C.

## Discussion

The resection of epileptogenic tissue in the mesial temporal lobe represents the most important factor in the success of temporal lobe surgery. A variety of different approaches have been developed to access these mesial structures in the dominant hemisphere, because language-bearing neocortical regions impede access to this region. Selective resections of the amygdala and hippocampus, via a variety of approaches, minimize the disturbance of temporal language regions. However, the access to the amygdala, hippocampus, and entorhinal cortex is necessarily more limited, and there is a concern that this limitation could lead to incomplete resections and poorer outcome. In this study, we directly compared the extent of mesial resections and seizure outcomes in an institutional cohort, finding that the type of surgical approach influenced neither of these outcomes. Resection of the amygdala, hippocampus, and entorhinal cortex was upward of 90%, regardless of the type of approach. Given this result, the finding that seizure outcomes were not significantly



**FIG. 3.** Graphs showing comparisons of neuropsychiatric parameters grouped by pre- and postoperative values. **A:** The BNT scores. **B:** The memory scores (VM, PM). **C:** The MMSE scores. **D:** The IQ scores (VIQ, PIQ, FSIQ). The error bars delimit the SEM.

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different between groups was not surprising. With either ATL or SelAH, 86%–90% of patients were seizure free at 2 years, comparing favorably with the expected range of outcomes reported in the literature.<sup>12,15,19</sup> The early neuropsychological and language sequelae of these surgeries reflected the lateralization of the resection, but in general did not correlate with seizure outcome. The only factor found to correlate with a poor seizure outcome was the use of invasive monitoring (subdural grids or depth electrodes). This is not surprising, since the reason for implantation of these intracranial electrodes was usually related to a question of a second ictal focus.

This study cannot answer the question of whether there is an extent of resection necessary for a good seizure outcome. Because the patients in this study had 90% or more of their mesial structures resected, it would be impossible to determine whether there is a correlation between a less extensive resection and a poorer seizure outcome. Similarly, this study cannot determine whether there is a correlation between specific neuropsychological test score changes and the extent of resection. Finally, we cannot comment on other techniques for SelAH, which use either a transsylvian route or a subtemporal approach. Nevertheless, it does appear that transcortical, image-guided SelAH affords excellent access to the mesial structures and a seizure outcome that is comparable to standard temporal lobe resection. Although these conclusions are limited by the retrospective nature of this study, they do support consideration of selective mesial resection, even in the nondominant temporal lobe.

### Conclusions

Selective amygdalohippocampectomy enables the surgeon to remove the amygdala, hippocampus, and entorhinal cortex slightly less completely than is normally achieved using a standard temporal lobectomy. Nevertheless, seizure outcomes and neuropsychological sequelae from the two surgical approaches appear to be comparable.

### Disclosure

The authors report no conflict of interest concerning the materials or methods used in this study or the findings specified in this paper.

Author contributions to the study and manuscript preparation include the following. Conception and design: Sagher. Acquisition of data: Thawani, Gomez-Hassan. Analysis and interpretation of data: all authors. Drafting the article: Sagher, Thawani, Etame. Critically revising the article: Sagher, Etame, Gomez-Hassan. Reviewed submitted version of manuscript: all authors. Approved the final version of the manuscript on behalf of all authors: Sagher. Statistical analysis: Thawani. Study supervision: Sagher.

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